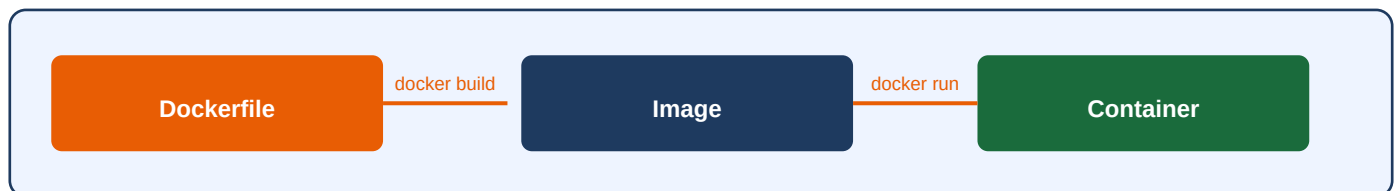


# DockerFundamentals

Dockerfile · Components · Build · Run

## How Docker Works



- **1. Dockerfile** — A plain text file containing instructions that tell Docker how to assemble the image.
- **2. Image** — The result of `docker build`. A read-only snapshot of a configured environment — like a template.
- **3. Container** — The result of `docker run`. The image brought to life as an isolated, running process.

### Example workflow:

```
# Step 1 — Build an image from the Dockerfile in current directory
docker build -t my-app:1.0 .

# Step 2 — Run a container from that image
docker run -d -p 8080:80 my-app:1.0
```

## Dockerfile Components — Quick Reference

| Instruction  | What It Does                                                                                           | Example                                                        |
|--------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>FROM</b>  | Specifies the base image. Every Dockerfile must start with FROM.                                       | FROM ubuntu<br>FROM nginx:latest                               |
| <b>LABEL</b> | Adds metadata (author, version, description) to the image.                                             | LABEL author="Mustafa"<br>LABEL version="1.0"                  |
| <b>COPY</b>  | Copies files/folders from your local machine into the container. Local paths only — cannot fetch URLs. | COPY app.py /app/<br>COPY ./src /app/src                       |
| <b>ADD</b>   | Like COPY, but also downloads from URLs and auto-extracts .tar/.gz archives.                           | ADD https://example.com/f.tar.gz /app/<br>ADD app.tar.gz /app/ |

| Instruction       | What It Does                                                                                                                                                               | Example                                                                                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>RUN</b>        | Executes a command during the image BUILD. Used to install packages, create directories, configure the environment.                                                        | <code>RUN apt-get install -y git</code><br><code>RUN yum install git -y</code>                      |
| <b>CMD</b>        | Default command when a container STARTS. Can be overridden by passing a command to docker run.                                                                             | <code>CMD ["python3", "app.py"]</code><br><code>CMD echo "Hello"</code>                             |
| <b>ENTRYPOINT</b> | Also runs at container start, but has higher priority than CMD and is NOT overridden by docker run arguments. When used with CMD, CMD provides default args to ENTRYPOINT. | <code>ENTRYPOINT ["python3"]</code><br><code>CMD ["app.py"]</code>                                  |
| <b>WORKDIR</b>    | Sets the working directory inside the container. Creates the directory if it doesn't exist. All subsequent instructions run from this path.                                | <code>WORKDIR /myapp</code><br><code># Like: mkdir /myapp &amp;&amp; cd /myapp</code>               |
| <b>ENV</b>        | Sets runtime environment variables inside the container. Values persist when the container runs and are accessible by the application.                                     | <code>ENV APP_PORT=8080</code><br><code>ENV DB_HOST=localhost</code>                                |
| <b>ARG</b>        | Defines build-time variables passed via --build-arg. NOT available inside the running container — only during the build.                                                   | <code>ARG VERSION=1.0</code><br><code>docker build --build-arg VERSION=2.0</code><br><code>.</code> |
| <b>EXPOSE</b>     | Documents which port the container listens on. Informational only — does not publish the port. Use -p in docker run to actually publish.                                   | <code>EXPOSE 8080</code><br><code>EXPOSE 443</code>                                                 |

## Key Distinctions to Remember

### COPY vs ADD

|                                                                                                                      |                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>COPY</b><br>Copies local files only. Simple and predictable.<br>■ Preferred for most cases — use COPY by default. | <b>ADD</b><br>Copies local files AND downloads from URLs AND extracts tar archives.<br>■■ Use only when you specifically need URL download or tar extraction. |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|

### RUN vs CMD vs ENTRYPOINT

|            |                                       |                                                                                             |
|------------|---------------------------------------|---------------------------------------------------------------------------------------------|
| <b>RUN</b> | <b>When:</b> Runs at IMAGE BUILD time | Install packages, create dirs, configure environment. Result is baked into the image layer. |
|------------|---------------------------------------|---------------------------------------------------------------------------------------------|

|                   |                                      |                                                                                                 |
|-------------------|--------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>CMD</b>        | <b>When:</b> Runs at CONTAINER START | Sets the default command. Overridden if you pass a command to docker run.                       |
| <b>ENTRYPOINT</b> | <b>When:</b> Runs at CONTAINER START | Higher priority than CMD. NOT overridden by docker run args. CMD becomes its default arguments. |

## ENV vs ARG

|                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ENV</b> <ul style="list-style-type: none"> <li>■ Available inside the running container</li> <li>■ Set once in Dockerfile, used by your app at runtime</li> <li>■ Cannot be changed with --build-arg</li> </ul> <i>Use for: port numbers, connection strings, app config</i> | <b>ARG</b> <ul style="list-style-type: none"> <li>■ NOT available inside the running container</li> <li>■ Can be overridden at build time with --build-arg</li> <li>■ Useful for parameterising the build itself</li> </ul> <i>Use for: version tags, build-time secrets, conditionals</i> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Sample Dockerfile — Putting It All Together

---

```
# Base image
FROM ubuntu:22.04

# Author metadata
LABEL author="Mustafa" version="1.0"

# Build-time variable (not available inside the running container)
ARG APP_VERSION=1.0

# Runtime environment variable (available inside the running container)
ENV APP_PORT=8080

# Install dependencies during image build
RUN apt-get update && apt-get install -y python3 git

# Set working directory – creates /myapp if it doesn't exist
WORKDIR /myapp

# Copy application code from local machine into the container
COPY app.py .

# Download and extract an external config archive
ADD https://example.com/config.tar.gz /myapp/config/

# Document the port this container listens on
EXPOSE 8080

# Fixed entry command – not overridden by docker run arguments
ENTRYPOINT ["python3"]

# Default argument to ENTRYPOINT – can be overridden
CMD ["app.py"]
```

### ■ Quick Memory Rules

**RUN** → build time (baked into image) | **CMD & ENTRYPOINT** → container start time

**COPY** → local files only | **ADD** → local + URLs + tar extraction

**ENV** → visible inside container | **ARG** → build time only, NOT in container

**WORKDIR** → creates & cd's into the directory automatically

**EXPOSE** → documentation only; use **-p** in docker run to actually publish the port